The Probe Is Part of the Process

Observation-induced change, causal confounding, and safe telemetry budgets in industrial edge intelligence

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Version: 1.0 **Publication date:** November 11, 2026

Publication state: Author-released working paper - not peer reviewed

DOI: Not assigned (GitHub edition)

THESIS / Observation is an intervention when its resource, physical, or policy footprint can change the system being observed. Its cost and causal role should be governed as evidence, not hidden in the telemetry plumbing.

Abstract

Industrial intelligence systems often treat observation as an external, low-cost operation. Yet a sensor mount can alter a structure's measured dynamics; an active network probe can affect a controller; a telemetry collector can compete for compute resources; and a change-triggered polling policy can make the observer respond to the very behavior it seeks to explain. These mechanisms differ: some change the physical or computational system, others distort the observation channel, and still others introduce selection bias. This paper argues that observation policy should be modeled as an explicit, bounded intervention rather than invisible infrastructure. It proposes an Observation Impact Contract (OIC): a claim-specific record of probe identity, modality, schedule, permitted operating regimes, resource and safety budgets, independent witness signals, causal uncertainty, and stop conditions. A causal model and a controlled, randomized switchback evaluation protocol are developed to distinguish probe-induced effects from regime drift and normal process variation. The proposal is grounded in OT safety guidance, SNMP and OPC UA protocol semantics, vibration-metrology standards, and experimental-design literature. It is an architectural argument and test plan, not evidence that any particular production system has suffered probe-induced faults. The proposed contract has not been implemented or empirically validated in this paper.

Keywords: observer effect; industrial telemetry; OT safety; active probing; causal inference; instrumentation overhead; edge intelligence; SNMP; OPC UA; adaptive sampling; switchback experiments; evidence governance.

Suggested citation: Yarian, M. (2026). The Probe Is Part of the Process: Observation-induced change, causal confounding, and safe telemetry budgets in industrial edge intelligence. SynapticSteel Technical Papers, STP-006, version 1.0. Yarian Works, LLC. DOI to be assigned upon publication.

> **Scope and evidence class**
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> This is a technical architecture and evaluation proposal. Protocol and safety claims are supported by the cited standards and literature; the proposed Observation Impact Contract and its predicted benefits are **not** validated outcomes. No live industrial plant was probed or modified for this paper.

1. The measurement boundary is not outside the system

Industrial analytics often represent the plant as a source of data and the observer as an external consumer. That separation is operationally convenient, but it is not universally causal. The act of acquiring information can consume controller time, occupy network buffers, contend for edge CPU cycles, change sensor loading, or alter when an anomaly is likely to be recorded. An alarm may therefore reflect a plant disturbance, an observation artifact, an observation-induced disturbance, or a mixture of all three.

The distinction matters most when an edge system increases its scrutiny after a suspected event. If a rising anomaly score causes more aggressive polling, and that polling consumes a scarce controller or compute resource, the observer has entered a feedback loop. Even when the probe does not affect the plant, event-triggered collection changes which states are densely sampled. Treating these extra observations as if they were a random sample of ordinary operation can bias downstream estimates.

This paper advances a narrow thesis: **an observation policy should have an explicit impact contract, with independently witnessed resource and safety limits, and its actions should be retained as potential causal inputs to later diagnosis.** It does not claim that all sensing is materially invasive. For many systems, observation cost is negligible. The claim is that negligibility is an empirical property of a particular modality, target, regime, and schedule, not an axiom.

NIST guidance already warns that active scans can destabilize sensitive OT devices and affect process state [1]. Experimental SCADA work documents the need to test scanner effects on specific equipment [2], while work at Pacific Northwest National Laboratory proposes combining passive discovery with carefully selected active tests [3]. The gap addressed here is broader than security scanning: it is the systematic treatment of routine telemetry, instrumentation, and adaptive sampling as first-class inputs to an industrial evidence model.

2. Four mechanisms that must not be conflated

2.1 Physical loading and transducer coupling

A contacting accelerometer and its mount form part of the mechanical measurement chain. ISO 5348:2021 describes how mounting conditions influence frequency response and measurement fidelity [4]. In a lightweight or compliant structure, the mount or cable can also change the coupled dynamics; in other situations the primary effect is measurement distortion rather than a consequential change to plant operation. These are different failure modes and require different reference measurements. Similar issues arise when thermal probes have finite response time or exchange heat with the measured medium [5].

2.2 Protocol and controller workload

A polling operation is a request for computation and communication, not merely a read of an abstract variable. SNMP GetBulk permits a requester to ask for many variable bindings in one operation, and RFC 3416 explicitly addresses response-size constraints and cases in which expensive requests may be truncated [6]. OPC UA subscriptions negotiate sampling and publishing behavior; the server may revise intervals, and queueing, filters, and delivery timing can affect what the client receives [7]. Neither protocol specification guarantees that a particular controller can serve an arbitrary request rate without process impact.

2.3 Edge-compute competition

Collection, decoding, encryption, logging, visualization, and inference share finite CPU, memory, I/O, and power budgets on an edge device. A collector can perturb the same host metrics it reports. This is an engineering inference from shared-resource scheduling, not a claim that a particular TelePod collector has demonstrated such an effect. NIST IR 8188 provides a precedent for measuring performance impacts of added cybersecurity functions in a process-control testbed [8].

2.4 Selection and representation bias

Observation may be distorted without altering the plant. Queue overflow, deadbands, negotiated sample rates, duplicated polls, and timestamp differences can change the apparent temporal structure of a signal. More subtly, adaptive collection makes observation intensity a function of earlier anomalies. In causal terms, the sampling policy becomes endogenous: the observer chooses when to look based on what it has already seen. More data after an alert is not automatically stronger evidence of the alert's cause.

Mechanism	Primary effect	Independent check
Mount / contact	Transfer-function distortion; possible structural loading	Reference accelerometer; mount-swap or shaker calibration
Active protocol request	Controller/network workload; possible process interference	PLC cycle time, request latency, packet capture
On-host telemetry	CPU / memory / power contention	Separate low-impact logger or hardware counter
Adaptive sampling	Unequal selection; temporal representation bias	Full probe schedule and regime-matched control windows

3. What established approaches solve, and what they miss

Passive monitoring minimizes direct target interaction and is often preferable for OT discovery [1,3]. Its blind spots include silent devices, encrypted traffic, missing local physical measurements, and dependence on existing traffic. Passive is a modality description, not a guarantee of zero system cost: the capture host and mirror infrastructure still consume resources.

Fixed-rate polling is simple, auditable, and often sufficient. It can, however, synchronize traffic bursts across devices, ignore operating regimes, and create a persistent load. A low mean bandwidth figure is not enough to establish safety if worst-case controller execution time or response-tail latency is the actual constraint.

Subscriptions and change-driven reporting can reduce unnecessary network exchanges. OPC UA separates sampling from publication and permits filters and queues [7]. Those mechanisms relocate and shape workload rather than making it disappear. The negotiated settings, overflow behavior, and server-side sampling cost must be recorded.

Active testing can reveal otherwise invisible state. PNNL's risk-optimized approach shows why selective probing is preferable to indiscriminate scanning [3]. However, a discovery framework does not by itself resolve whether the act of observation contaminates subsequent anomaly attribution.

Digital twins and residual models can compare measured behavior against predicted behavior. They do not automatically distinguish a genuine plant change from a probe-induced change when the model omits the probe policy. Adding observer action to the causal record is complementary to, not a replacement for, model-based diagnostics.

4. A causal model for observer-induced change

Let **U(t)** denote operating regime and unobserved disturbances, **A(t)** the probe schedule or intensity, **L(t)** resource/physical load attributable to observation, **Y(t)** the plant or target state, and **X(t)** the recorded telemetry. A useful structural decomposition is:

Conceptual structural equations

$L(t) = g(A(t), U(t), \text{device capability}) + e_L$
$Y(t) = f(Y(t-1), U(t), L(t)) + e_Y$
$X(t) = h(Y(t), A(t), \text{queue/filter state}) + e_X$
$A(t) = \text{policy}(H(t-1))$, where H is the observation history. These are modeling assumptions, not fitted equations or empirical results.

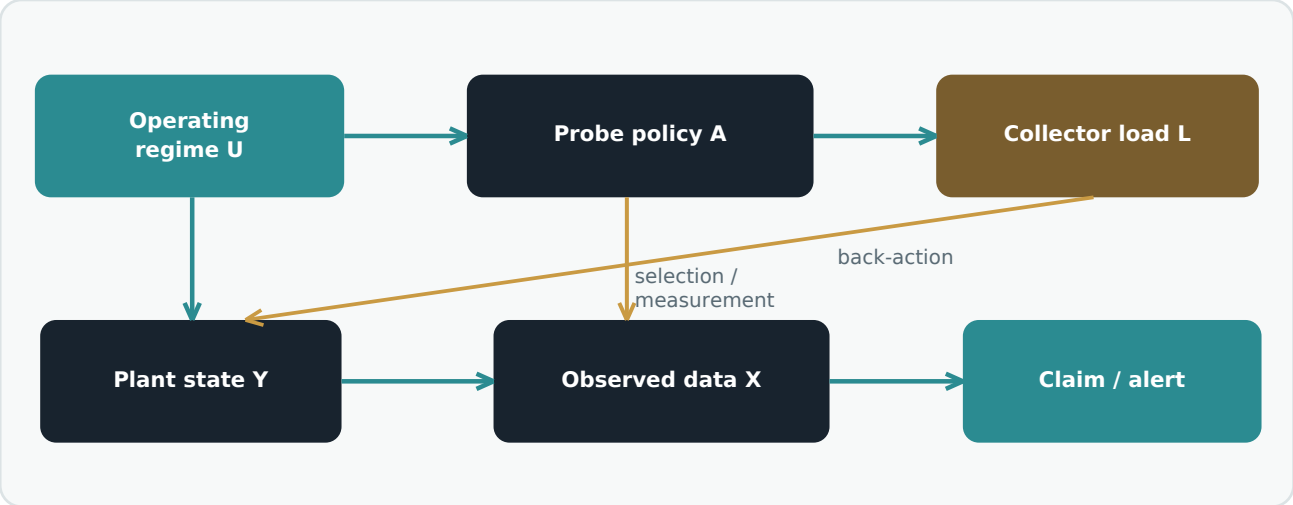


Figure 1. Simplified causal structure. The observer can affect target state through load, affect recorded data directly through sampling, and respond to regime or previous observations. The graph is explanatory, not an identified causal model.

The relevant estimand is not simply the correlation between poll rate and alarms. It is a regime-specific contrast in target behavior under two permissible observation policies, with prior probe history held or modeled appropriately:

Proposed causal estimand

Delta(r, k) = E[Y(t) | do(A(t-k:t) = high), U(pre) in r] - E[Y(t) | do(A(t-k:t) = low), U(pre) in r]
The do notation is a target causal contrast. Estimation requires defensible assignment, measurement, and carryover assumptions; the expression alone identifies nothing.

The observation channel also needs its own estimand: the difference in measurement fidelity or missingness under the same two policies. Without an independent witness, a change in X cannot be confidently partitioned into a change in Y and a change in measurement. This is an identifiability limit, not a software defect.

5. The proposed Observation Impact Contract

The Observation Impact Contract (OIC) is a **proposal** for a machine-readable record attached to an observation source and its operating policy. It is not a universal scalar "probe safety score." Its purpose is to preserve which questions a probe can answer, under which regimes, with what resource budget and evidence of harmlessness, and who authorized the policy.

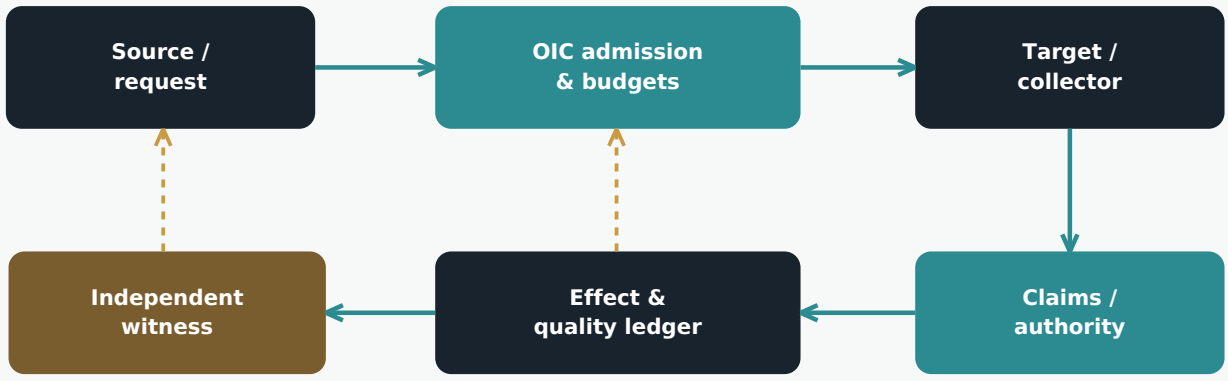


Figure 2. Proposed OIC architecture. Probe requests pass a bounded admission gate; independent witness measurements and probe-action history enter a ledger. The ledger informs claim confidence, but cannot grant machine-control authority.

Contract field	Required content / reason
Identity & provenance	Source, target class, firmware/protocol, mount, collector version, owner
Action semantics	Passive tap, subscription, poll, diagnostic test, physical stimulus; requested and actual schedule
Regime & validity	Allowed modes, maintenance windows, known exclusions, expiry / revalidation triggers
Resource envelope	Rate, concurrency, bursts, timeouts, CPU/network/power constraints, priority
Witness & uncertainty	Independent signals, calibration, missingness, effect-size interval, residual confounding
Authority & stop	Approver, target safety classification, abort thresholds, rollback and evidence retention

Admission is claim-specific. A passive network tap may be adequate for protocol traffic attribution but insufficient to infer shaft vibration. A host-level CPU counter may be useful for resource contention but unsuitable as its own sole witness of collection overhead. An OIC must record what was observed, what action made observation possible, and what the evidence cannot support.

6. Budgeting without inventing a universal safe polling rate

An implementation can begin with a per-target token bucket for request admission and a stricter cap on concurrent requests. However, tokens based only on bytes or requests per second are insufficient. Device-specific service time, peak burst cost, response tails, and process deadlines matter. The budget must therefore combine a configured limit with observed stress indicators and a conservative abort rule.

For a target d, define a proposed normalized demand index over window W as $\mathbf{C_{d,W} = sum_{i\ in\ W}\ w(type_i, size_i, response_i, regime)}$. The weights are **unknown until measured or conservatively bounded**; this is a scheduling abstraction, not a calibrated safety metric. Separate hard ceilings should constrain concurrency, burst length, retry amplification, and worst-case latency. A failed read must not automatically trigger an unbounded retry storm.

A safe system should fail toward lower observation intensity or passive evidence when a limit is exceeded, while clearly marking the resulting loss of coverage. If an alarm occurs during a probe burst, the probe schedule is a candidate explanatory variable. If the witness signal is missing, the system should report "effect unknown," not "probe harmless."

> **A critical separation**
>
> **Evidence admission is not execution authority.** The OIC can govern whether a read or diagnostic observation is admissible under an approved policy. It does not authorize actuator commands, PLC program changes, process setpoint changes, or unsafe active scans. Those require independent plant-owner authority and safety controls.

7. A falsifiable evaluation protocol

A meaningful test must separate natural regime changes, the measurement channel, and the observer's intervention. Randomized switchback experiments offer one possible design when a single test unit must alternate between observation policies and carryover is modeled. Bojinov, Simchi-Levi, and Zhao derive design and inference methods for switchback settings with treatment persistence [9]. Their application domain is not industrial safety; applying the method here requires additional hazard review and instrumentation validation.

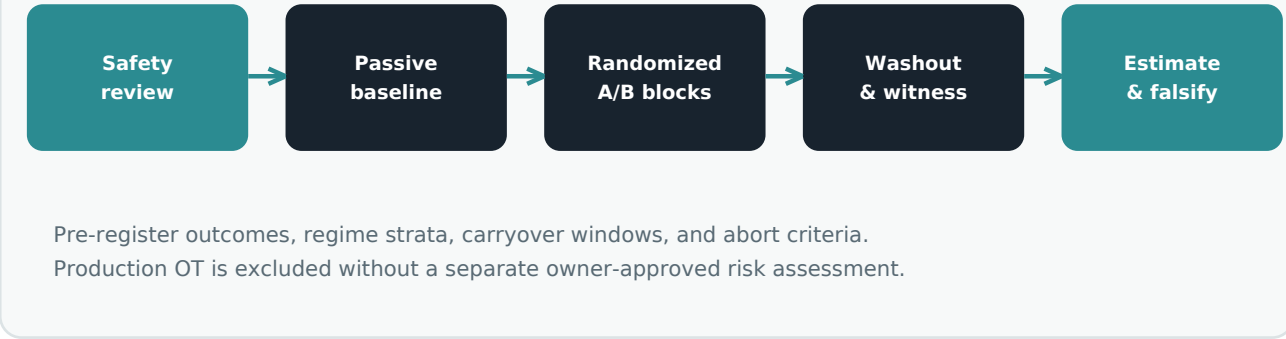


Figure 3. Laboratory evaluation sequence. Randomization applies only to preapproved low-risk probe policies on a controlled testbed, not to production plant interventions.

7.1 Preconditions and design

• **Hazard review first:** classify target criticality, identify control-loop deadlines, and exclude any policy with unbounded or poorly characterized effects. Never discover a safe rate by stressing a live production PLC.

• **Independent witness:** capture at least one outcome through a channel whose workload is not controlled by the candidate probe policy. Validate the witness's own loading and clock relationship.

• **Regime blocking:** stratify by load, shift, equipment mode, thermal state, or other predeclared operating conditions. Keep configuration and maintenance epochs explicit.

• **Randomized switchback:** alternate only among preapproved low-impact observation policies, with randomized order, sufficient washout, and predeclared carryover windows. Record assignment and deviations.

• **Pre-register endpoints:** target scan-cycle jitter or service latency, response-tail latency, missed data, host CPU/power, and physically relevant process metrics. Use effect-size intervals, not only p-values.

• **Abort and audit:** stop on a hard safety threshold, retain the complete action schedule and raw evidence, and distinguish protocol failure from a measured plant effect.

7.2 What would count as evidence?

The primary hypothesis is that, for some target-regime pairs, increasing observation intensity changes at least one independently measured performance or physical outcome by more than a predeclared practical-effect threshold. A valid null result requires uncertainty bounds sufficiently narrow to rule out that threshold under tested conditions; a non-significant p-value alone is not proof of harmlessness. Conversely, a positive difference without randomized or otherwise credible assignment may reflect confounding, especially when probing is triggered by existing anomalies.

Negative controls help detect instrumentation artifacts: compare a metric that should not respond to the probe; vary probe timing without changing expected total work; and inspect whether anomalies align with collector scheduling rather than process physics. These checks can falsify simple

explanations but cannot eliminate all unmeasured confounding.

7.3 Analysis and reporting

Report intention-to-treat contrasts by randomized block, treatment history, regime, and target class. Estimate confidence intervals with methods that respect serial dependence and the assignment design [9]. Publish both target-outcome effects and observation-quality effects, including missingness, delayed delivery, and sample-selection changes. Predeclare multiplicity handling for multiple channels and separate exploratory findings from confirmatory endpoints.

8. Counterarguments and limitations

"Most monitoring is harmless." Often true. The proposed contract should be lightweight when the effect has been shown negligible for a bounded context. It should not impose research-grade experiments on every ordinary sensor read. The burden of proof scales with criticality and plausible back-action.

"A second sensor can also disturb the plant." Correct. Independent does not mean zero-impact. Witness independence is an engineering claim about failure modes and intervention pathways, not merely a different vendor label. Its limitations belong in the contract.

"Passive monitoring solves the problem." It reduces direct interaction with a target but does not cover all physical variables, silent devices, or local compute effects. It is an important baseline and sometimes the correct final architecture, not a complete substitute for measurement-specific analysis.

"Randomization is unsafe or impractical." On production systems, this objection can be decisive. In such cases the correct outcome is a narrower observational claim, a laboratory surrogate, or a decision not to test. A causal estimate is not owed to the architecture if obtaining it would endanger operations.

"The OIC adds bureaucracy and overhead." It does. Its value must be measured against simpler per-target rate limits, static allowlists, and existing OT change-control procedures. A proposed implementation should be rejected if it adds complexity without reducing false attribution, safety risk, or debugging time.

"Causal identification remains incomplete." Yes. Hidden plant disturbances, nonlinear feedback, long carryover, mount effects, and insufficient independent measurements can prevent clean identification. The OIC makes these unknowns visible; it does not make them disappear.

9. Experimental scope and implementation boundary

This paper is a research proposal and evaluation protocol. Private operational checkpoints are excluded from this public edition. No implementation, deployment, field validation, or measured performance benefit is established by the proposal. A prototype should first be evaluated with recorded or simulated inputs, explicit baselines, failure cases, and independently measured outcomes.

10. Research agenda and disconfirmation criteria

The proposal is useful only if it survives empirical comparison with simpler approaches. Three preregistered questions should lead a follow-on study:

- **Impact:** Does the candidate probe policy produce a practically meaningful change in an independent target outcome within at least one approved regime?
- **Attribution:** Does retaining probe-action history improve discrimination between endogenous process anomalies and observer-linked anomalies over a baseline model that ignores probe history?
- **Operational value:** Does OIC-based admission reduce unsafe load, missed evidence, or false attribution compared with a fixed-rate budget and ordinary OT change control, without unacceptable coverage loss?

The thesis would be weakened if well-powered tests repeatedly bound all relevant probe effects below practical thresholds, if probe-history features fail to improve out-of-sample attribution, or if OIC governance adds overhead without operational benefit. A credible result should publish the nulls, ablations, failure cases, and limits of transfer to other controllers.

Publication pathway: This paper is suitable for a technical working-paper / SSRN-style research discussion after editorial review. An arXiv or peer-reviewed research submission would require a reproducible protocol, sufficiently independent instrumentation, comparative baselines, safety approval, and empirical results. The present paper makes no priority or novelty claim beyond its explicitly proposed synthesis.

11. Conclusion

Industrial intelligence cannot assume that observation lives outside the system. The sensor, collector, protocol, and sampling policy may affect what happens, what is recorded, or which moments become visible. A rigorous system should distinguish those mechanisms, retain the observer's own action history, and demand independent evidence before treating observation as causally neutral. The proposed Observation Impact Contract is one way to make that discipline explicit. Its merit will be established not by architecture diagrams but by falsifiable, bounded tests against simpler monitoring policies.

Related SynapticSteel Technical Papers

STP-001 - Failure Metabolism. Related through evidence-preserving operational history and the treatment of interventions as context. A Zenodo working-paper record has been documented at DOI 10.5281/zenodo.23130549; it is not a peer-reviewed validation of STP-006.

STP-005 - Trust Is Not Agreement. Related through the separation of channel credibility, consensus, and claim-specific admission. Its DOI was reserved during preparation of this paper, but its final public release was not independently confirmed here; no public DOI citation is asserted.

References

[1] Stouffer, K., et al. (2023). Guide to Operational Technology (OT) Security. NIST SP 800-82 Rev. 3. DOI: 10.6028/NIST.SP.800-82r3. Source

[2] Coffey, K., Smith, R., Maglaras, L., & Janicke, H. (2018). Vulnerability Analysis of Network Scanning on SCADA Systems. Security and Communication Networks, 2018, 3794603. DOI: 10.1155/2018/3794603. Source

[3] Edgar, T. W., Niddodi, S., Rice, T. R., et al. (2019). Safer and optimized vulnerability scanning for operational technology through integrated and automated passive monitoring and active scanning. Journal of Information Warfare, 18(4), 125-155. PNNL-SA-145717. Source

[4] International Organization for Standardization. (2021). ISO 5348:2021, Mechanical vibration and shock - Mechanical mounting of accelerometers. Source

[5] Pitts, W. M., Braun, E., Peacock, R. D., et al. (2001). Temperature uncertainties for bare-bead and aspirated thermocouple measurements in fire environments. ASTM technical proceedings / NIST publication record. Source

[6] Presuhn, R. (Ed.). (2002). Version 2 of the Protocol Operations for the Simple Network Management Protocol (SNMP). RFC 3416, IETF. Source

[7] OPC Foundation. (accessed 2026-10-09). OPC Unified Architecture, Part 4: Services, sections 5.13 and 5.14 (MonitoredItem and Subscription service sets). Source

[8] Tang, C. (2017). Key Performance Indicators for Process Control System Cybersecurity Performance Analysis. NIST IR 8188. DOI: 10.6028/NIST.IR.8188. Source

[9] Bojinov, I., Simchi-Levi, D., & Zhao, J. (2023). Design and Analysis of Switchback Experiments. Management Science, 69(7), 3759-3777. DOI: 10.1287/mnsc.2022.4583. Source

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